

Part 6

Inferences about more than two population central values
Analysis of Variance and F-test
Completely Randomized Design

Inference about more than Two Population Central Values

- In Part 3 we studied inferences about a population mean μ using a single sample from the population.
- In Part 4 we studied inferences about the difference between two population means $\mu_1 - \mu_2$ based on a random sample from each population.
- Part 6 deals with inferences about means $\mu_1, \mu_2, \dots, \mu_t$ from $t > 2$ populations.
- The first and foremost question that we ask is whether the populations have different means?

Inference about more than Two Population Central Values - cont.

- A natural approach to think of when faced with the task of determining whether evidence points to $\mu_1 = \mu_2 = \mu_3 = \cdots = \mu_t$ or not, is to draw on knowledge acquired in Part 4 and perform t -tests of $H_0 : \mu_i = \mu_j$ for all pairs of means.
- If **all pairwise tests of means** fail to reject the null hypotheses, then one may conclude that all means μ_i are equal.
- This procedure has a serious flaw which will now be described.
- Take the case $t = 3$, so that we have three populations with means μ_1, μ_2, μ_3 .
- A sample from each population yields sample means and variances $\bar{y}_1, \bar{y}_2, \bar{y}_3, s_1^2, s_2^2, s_3^2$.

Inference about more than Two Population Central Values - cont.

- All possible differences $\mu_i - \mu_j$ are:

$$\mu_1 - \mu_2, \mu_1 - \mu_3, \mu_2 - \mu_3 .$$

- To test each $H_0 : \mu_i - \mu_j = 0$, a t -statistic used is of the form

$$t_{ij} = \frac{\bar{y}_i - \bar{y}_j}{s_p \sqrt{\frac{1}{n_i} + \frac{1}{n_j}}}, \quad s_p^2 = \frac{(n_i - 1)s_i^2 + (n_j - 1)s_j^2}{n_i + n_j - 2}$$

- Suppose each test is made at the same level α . Thus for each pair (i, j)

$$P(|t_{ij}| > t_{\alpha/2} \mid \text{when } H_0 \text{ is true}) = \alpha.$$

Inference about more than Two Population Central Values - cont.

- In other words, the Type I error probability for each test is α . We shall call α the per-comparison error rate.
- Now let us ask the question – What is the probability of making one or more Type I errors when we make all three tests? (We shall call this the overall error rate.)
- Theory says that if the three t random variables were statistically independent then the overall error rate is $1 - (1 - \alpha)^3$ [this works out to 0.14 if $\alpha = 0.05$, much larger than .05]
- The three test statistics are not independent. For example t_{12} and t_{13} both involve $\bar{y}_1$. Furthermore, all three have the same denominator. So they are not statistically independent.

Inference about more than Two Population Central Values - cont.

- Thus the overall error rate is not exactly 0.14 for our three test situation, and computing its exact value is difficult in theory.
- In general, for c tests made at level α , the overall error rate is larger than $1 - (1 - \alpha)^c$.
- This is the flaw in the multiple testing approach to testing $H_0 : \mu_1 = \mu_2 = \cdots = \mu_t$.
- One solution is to use a very small α for each test in order to have a reasonably small overall error rate.
- For example, in the previous situation, if we had $\alpha = .01$ then the overall error rate would be less than .03.

Inference about more than Two Population Central Values - cont.

- We would like to avoid doing this if possible, because we will lose power by using small α for each test.
- The alternative to making multiple t -tests is to make a single F -test of $H_0 : \mu_1 = \mu_2 = \cdots = \mu_t$ versus $H_a : \text{at least one } \mu_j \text{ is different in value.}$
- This is called the analysis of variance F -test.

The Analysis of Variance and the F -test

- We can, at this point, begin talking in terms of experiments and their outcome. An experiment is a planned data collection activity.
- A simplistic view is that an experiment is a planned way to observe the effect of **treatments** that are applied to **experimental units**.

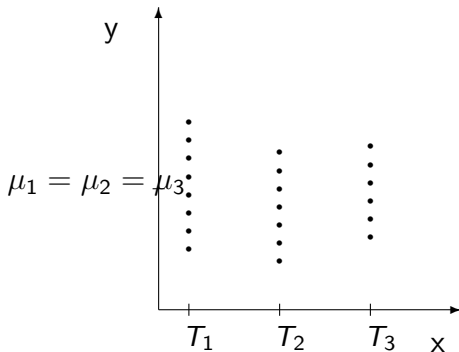
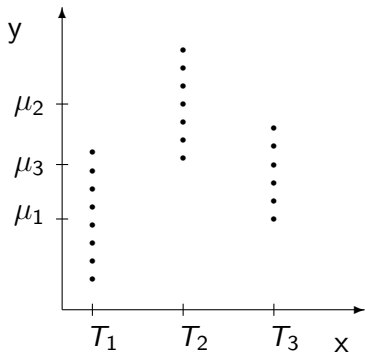
Examples

- Different amounts of a headache drug (treatments) are given to people with headache (experimental units) to observe the effect.
- Different furnace temperatures (treatments) are used to temper steel rods (experimental units) to see the extent of tempering for each temperature.

- In this context we can imagine theoretically a **parent population of experimental units** whose measure of interest has mean μ and variance σ_{ϵ}^2 .
- We imagine applying each treatment i to all elements in the parent population to obtain a **treated population**, say T_i .
- We will assume the treated population T_i has mean μ_i , possibly different than μ , but its variance is σ_{ϵ}^2 , i.e., the treatment doesn't affect the variance among experimental units, σ_{ϵ}^2 .

- In the past we have talked about populations without worrying about how they might have materialized. Here we are simply saying that we can imagine some populations as arising through experimentation.
- The simplest experimental design is named the **completely randomized design** (CRD).
- Here we have (say) t treatments and (say) nt experimental units. These experimental units are divided into t sets of n and the sets are assigned **at random**, each to a treatment.
- We can, equivalently, characterize this design as one that acquires a simple random sample of size n from each of t different treated populations $T_1, T_2, \dots, T_t$.

- Each of these populations has mean μ_i and variance σ_{ϵ}^2 , $i = 1, 2, \dots, t$.
- If all of the treatments have the population mean μ then $\mu_1 = \mu_2 = \dots = \mu_t = \mu$.
- If this were the case, **the variation among n samples** taken from each treated population $T_1, T_2, \dots, T_t$ (called the within sample variance) would be the same as the variation among all nt sample values.
- If some of the μ_j are different, then **the variance in the overall sample** will be inflated due to differences in population means, while the within sample variances would remain the same for each sample.



- This suggests that to investigate whether $\mu_1 = \mu_2 = \dots = \mu_t$ we may wish to compare the observed variation within samples from the populations T_i to the variation among (or between) samples.

- This is the motivation for analyzing variances when looking for differences in population means.
- Note that the **expected value of the mean square** among treatment groups is σ_ϵ^2 when $\mu_1 = \mu_2 = \dots = \mu_t$.
- This expectation gets larger than σ_ϵ^2 as the μ_i 's get further apart in value.
- Thus the ratio:

$$\frac{\text{Square between treatment groups}}{\text{Mean Square within treatment groups}}$$

has expected (average) value 1 when $\mu_1 = \mu_2 = \dots = \mu_t$ and greater than one otherwise.

- It will be difficult for us to decide how much larger than 1 the ratio must be before we declare that at least one μ_i is different from the others.
- The solution to this problem is to **assume the treated populations are Normally distributed**.
- Then the above ratio is an F random variable whenever $\mu_1 = \mu_2 = \cdots = \mu_t$.
- When the μ_i are not all equal, the distribution of this ratio will be shifted to the right.
- Thus the observed value of the above ratio will be larger than the **percentile value from the F-table** at a specified level α .

- Thus we may test the hypothesis

$$H_0 : \mu_1 = \mu_2 = \cdots \mu_t \text{ vs. } H_a : \text{at least one } \mu_i \text{ different,}$$

using the statistic $F = \frac{\text{Mean Square Between Groups}}{\text{Mean Square Within Groups}}$

- Since our test involves essentially all mean values $\bar{y}$'s, the central limit theorem tells us that we will have approximately an F random variable if the treated populations are not dramatically different than normal populations.
- The F-test is carried out by constructing an analysis of variance table discussed below.
- Handout gives details about computation of a relevant analysis of variance table for this case. Here we summarize the results.

The Analysis of Variance (AOV) Table for a CRD

- **Data:** (Note: the sample sizes in the t treatment groups need not be the same.)

$$\begin{array}{lcl} \text{Treatment 1:} & y_{11}, y_{12}, \dots, y_{1n_1} & \left| \bar{y}_1. \right. \\ \text{Treatment 2:} & y_{21}, y_{22}, \dots, y_{2n_2} & \left| \bar{y}_2. \right. \\ & \vdots & \left| \vdots \right. \\ \text{Treatment } t: & y_{t1}, y_{t2}, \dots, y_{tn_t} & \left| \bar{y}_t. \right. \end{array}$$

Let $n_T = n_1 + n_2 + \dots + n_t$ denote total number of observations.

The Analysis of Variance (AOV) Table for a CRD

- The **AOV table** in this case is:

Source	df	SS	MS
Treatment	$t - 1$	$SSB = \sum_{i=1}^t n_i (\bar{y}_{i.} - \bar{y}_{..})^2$	$SSB / (t - 1)$
Error	$n_T - t$	$SSW = \sum_{i=1}^t \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i.})^2$	$SSW / (n_T - t) \equiv \text{MSE}$
Total	$n_T - 1$	$\sum_i \sum_j (y_{ij} - \bar{y}_{..})^2$	

- where

$$SSW = \sum_i^t \sum_j^{n_i} (y_{ij} - \bar{y}_{i.})^2 = (n_1 - 1)s_1^2 + (n_2 - 1)s_2^2 + \cdots + (n_t - 1)s_t^2$$

$$\text{MSE} = SSW / [(n_1 - 1) + (n_2 - 1) + \cdots + (n_t - 1)] = s^2$$

- SSW is simply the familiar **pooled SS** that we saw for the 2 sample case back in Chapter 6, extended to the $t > 2$ case.
- MSE is the **pooled estimator of the population variance**, σ_ϵ^2 .
- **Model for data from a CRD**

The model of the CRD which we will use is:

$$y_{ij} = \mu + \alpha_i + \epsilon_{ij}, \quad i = 1, 2, \dots, t; \quad j = 1, 2, \dots, n$$

where $E(\epsilon_{ij}) = 0$, $Var(\epsilon_{ij}) = \sigma_\epsilon^2$, and the ϵ_{ij} 's are all independent.

- Here α_i is the **effect** due to treatment i .

- Thus the treated population means are $\mu_i = \mu + \alpha_i$ in this notation.
- If none of the treatments has an effect then $\alpha_1, \alpha_2, \dots, \alpha_t$ are all zero which is equivalent to $\mu_1 = \mu_2 = \dots = \mu_t = \mu$ for some μ .
- We may talk of **treatment means** or **treatment effects** in the single treatment factor CRD experiments irrespective of sample size.

Example 8.1 - Handout

- Data:

A horticulturist was investigating the phosphorous content of tree leaves from three different varieties of apple trees(1,2, and 3). Random samples of five leaves from each of the three varieties were analyzed for phosphorous content. The data are:

Variety	Phosphorous Content					Sample Size	Sample Mean	Sample Variance
1	.35	.40	.58	.50	.47	5	0.460	.00795
2	.65	.70	.90	.84	.79	5	0.776	.01033
3	.60	.80	.75	.73	.66	5	0.798	.00617

Example 8.1 - Handout - cont.

- Analysis of Variance Table:

Source of Variation	d.f.	S.S.	M.S.	F
Variety	2	.277	.138	17.25
Error	12	.0978	.008	
Total	14	.3748		

- Since the computed F value exceeds $F_{.05,2,12} = 3.89$, the null hypothesis that the mean phosphorus content for the three varieties are all equal is rejected.
- It appears that the mean phosphorus content for Variety 1 is smaller than those for Varieties 2 and 3. Techniques for testing such hypotheses will be discussed in the Chapter 9.

Checking the Equal Variance Assumption

- To summarize, our model for the multiple population case is (possibly unequal sample size)

$$y_{ij} = \mu + \alpha_i + \epsilon_j \quad i = 1, 2, \dots, t; \quad j = 1, 2, \dots, n_i$$

where ϵ_{ij} 's are independent, normally distributed and for each i the population parameters are $(\mu_i, \sigma_{\epsilon}^2)$.

- A problem occurs if there is heterogeneity of variance.

$$H_0 : \sigma_1^2 = \sigma_2^2 = \dots = \sigma_t^2 \text{ vs. } H_a : \text{ not all } \sigma_i^2\text{'s are equal}$$

- A test of this hypothesis, proposed by Hartley (1940) was discussed in Chapter 7. The test is the F -max test

$$F_{\max} = \frac{\max(S_i^2)}{\min(S_i^2)}$$

- This uses the largest and smallest of the population sample variances.
- The test statistic presumes equal sample sizes. Recall that Table 12 gives critical values, for $\alpha = 0.05$ and $\alpha = 0.01$ for $df = n - 1$.
- If sample sizes are not equal use the largest n_i .

- When sample sizes are nearly equal, heterogeneity of variance is not as great a problem unless variances are severely different. Such cases are usually detectable by simply looking at the sample variances.
- Hartley's test is very sensitive to departures from normality. An **alternative test, Levine's test** was also briefly discussed in a handout.
- When heterogeneity appears to be present, a **variance stabilizing transformation** can sometimes be found.
- Look at handout for variance stabilizing transformations.
- Finding a useful variance stabilizing transformation is not, in general, easy.

- If an approximate relationship between σ^2 and μ is found by examining the sample means $\bar{y}_i$'s and the corresponding sample variances S_i^2 , then it is possible to determine an appropriate transformation using theoretical arguments.
- For example, if this relationship is of the form $\sigma^2 = k\mu$ for some constant k , then the transformations $y_T = \sqrt{y}$ is suggested.
- Or, if the relationship is of the form $\sigma^2 = k\mu^2$ for some constant k , then the transformation $y_T = \log(y + 1)$ is recommended. (See Table 8.15 for other possibilities).
- Then the transformed data are analyzed using the usual method.
See Example 8.4.